

## Feature Extraction Workshop

1. In this workshop, we shall perform Feature Extraction using PCA.
2. Please use the dataset **wine.csv** for the questions that follow.
3. Assume that our **label** is 'Cultivar' and the rest of the columns are our **features**.
4. Visualize your data by creating **distribution plots** for the 4 features - Alacidity of ash, Flavanoids, Hue and Proline.
  - a. Use Seaborn's histplot() function to generate the plots.
  - b. Differentiate each plot by the 'Cultivar' feature by setting hue='Cultivar' in the histplot() function.
5. Extract 4 new features (feature-extraction) from the dataset using PCA.
  - a. Print out the amount of variance captured for each Principal Component.
  - b. Print out the total amount of variance captured for the 4 Principal Components.
  - c. Print out the minimum number of Principal Components required to capture at least 85% of the information of the dataset.
  - d. State the maximum number of Principal Components that can be generated from our dataset?

### References

- histplot() reference - <https://seaborn.pydata.org/generated/seaborn.histplot.html>